

Jacob Bradley

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Skills

- **Machine Learning & AI:** Supervised & Unsupervised Learning, Deep Learning, Model Optimization, Recommendation Systems.
 - **Programming & Analytics:** Python, R, Julia, SQL, NumPy, Pandas, Scikit-learn, Spark MLlib.
 - **Frameworks & Tools:** TensorFlow, Keras, PyTorch, ONNX, Git, Docker, AWS (EC2, S3, Lambda), REST APIs.
 - **Data Science & Engineering:** Data Preprocessing, Feature Engineering, Statistical Modeling, ETL/ELT, Big Data Pipelines.
 - **Software Engineering:** Algorithm Design, API Development, High-Performance Computing (HPC), SLURM, Linux Systems.
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Professional Experience

Data Scientist Intern

BSA Healthcare | Remote | Feb 2023 – May 2025

- Designed and executed SQL queries and Python pipelines to prepare, validate, and standardize multi-source healthcare datasets for population health and clinical outcomes reporting.
- Applied supervised and unsupervised ML models (classification, clustering) to patient data, improving risk stratification prediction accuracy by 18%.
- Built ETL workflows integrating structured and unstructured data, reducing preparation time by 40%.
- Developed production-ready ML models in Scikit-learn and TensorFlow, enabling scalable analytics and decision-support solutions.

Graduate Software Engineer Assistant – Backend Architecture & Data Engineering

Midwestern State University | Wichita Falls | May 2024- May 2025

- Optimized SQL Server and PostgreSQL databases, achieving 80% faster query performance through indexing and execution plan tuning.
 - Trained deep learning models (TensorFlow, PyTorch) for biological classification, reducing misclassification rates by 25%.
 - Automated HPC workloads with SLURM job scripts and Bash, cutting runtime overhead by 40%.
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Projects

NeuroBeats- Biometric Music Recommendation Platform

- Developed a real-time ML recommendation engine using biometric data (heart rate, temperature) to predict mood and curate music playlists.
- Built backend with FastAPI + PostgreSQL, integrated with Spotify API, and deployed on a React Native frontend.
- Enhanced the existing model which improved personalization, enabling seamless playlist queueing and mood detection.

INS-Centered Protein Interaction Network

- Constructed a multi-layer protein-protein interaction (PPI) network from BioGRID/STRING datasets.
 - Applied graph algorithms (shortest path, centrality, community detection) in NetworkX, exporting annotated graphs to Cytoscape for visualization.
 - Delivered insights into insulin signaling pathways with applications in diabetes research.
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Education

M.S. in Computer Science

Midwestern State University

B.S. in Biology, Minor in Chemistry

Midwestern State University
